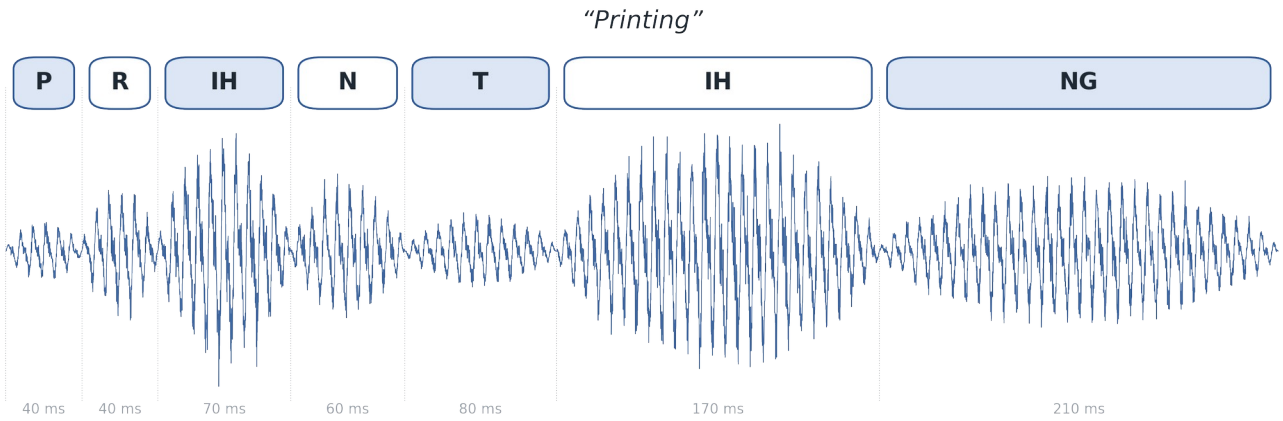
ADVANCED MACHINE LEARNING & DEEP LEARNING

Phonetic Guided Speech Generation

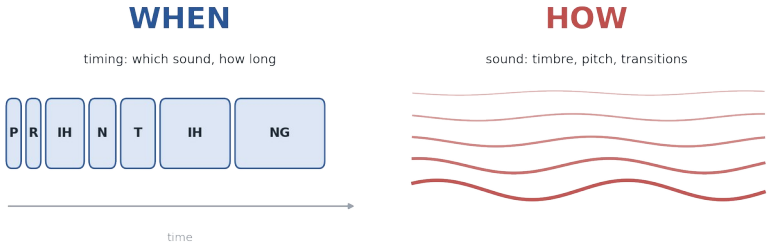
Explicit phoneme alignment and learned acoustic generation for text-to-speech



Shivam Mittal · Khushi Batra · Divya Singh · Sanath Waraikar · Aryan

Newton School of Technology, Rishihood University · github.com/shiavm006/phonetic-guided-tts

Speech has two parts: when, and how



If the timing is known exactly, how much of speech comes for free?

H1 Explicit timing alone cannot give intelligible speech; a learned acoustic model is needed.

H2 Explicit duration and pause models improve a neural decoder over its own timing.

H3 Fine-tuning the decoder on our alignments helps it render raw alignment timing.

Scope: diffusion, optional in the brief, is left as a motivated extension (see slide 12).

Where this work sits

Forced alignment

HMM-GMM aligners place phoneme boundaries: Montreal Forced Aligner [1]; statistical parametric TTS [2]

Neural TTS

Implicit alignment by attention: Tacotron 2 [3]. Explicit durations: FastSpeech 2 [4]

Generation

Neural vocoder HiFi-GAN [5]; diffusion decoder Grad-TTS [6] against over-smoothing

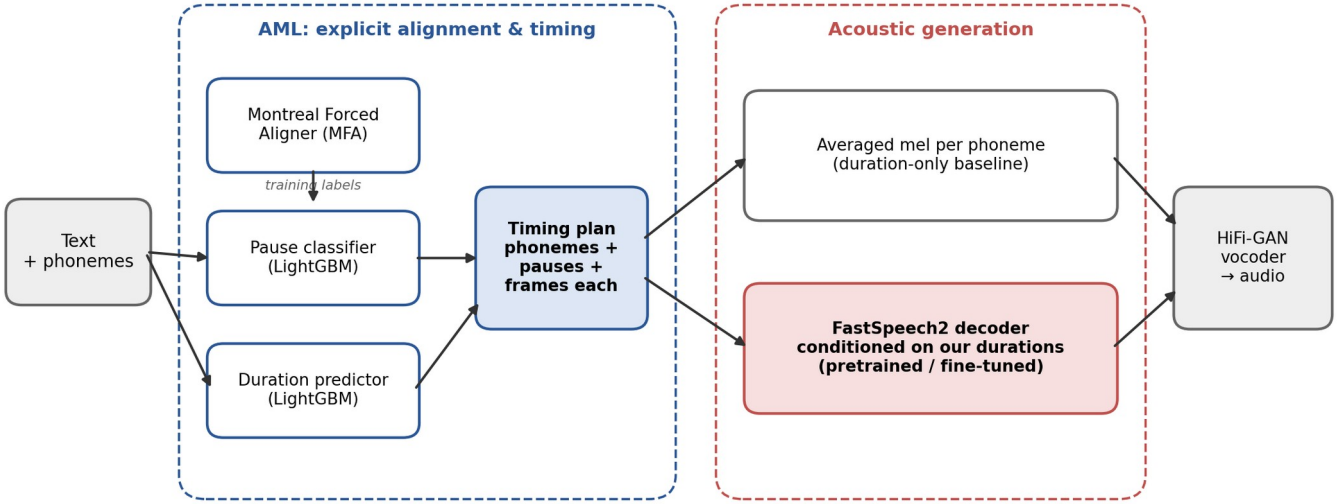
Evaluation

Held-out ASR for intelligibility: Whisper [7]; learned MOS predictor: UTMOS [8]

Our angle: hold one part fixed, change the other, on a leak-free test set.

[1] McAuliffe 2017 · [2] Zen 2009 · [3] Shen 2018 · [4] Ren 2021 · [5] Kong 2020 · [6] Popov 2021 · [7] Radford 2023 · [8] Saeki 2022. Full references in the report.

Classical timing, neural sound



LJSpeech 1.1 · 13,100 clips, 23.9 h, one speaker · MFA alignment, 0 failures · 80-band mels · 11,790 / 1,310 split by sentence

Two classical models predict the timing

DURATION · MAE ON SPEECH SOUNDS

MODEL	MAE
Per-phoneme mean	28.8 ms
Ridge (linear)	25.1 ms
MLP	19.3 ms
Random forest	18.8 ms
LightGBM	18.3 ms

PAUSE PLACEMENT · F1

MODEL	F1
Punctuation rule	58.7%
Logistic regression	61.1%
Random forest	63.0%
MLP	63.5%
LightGBM	64.2%

Linear models fall 7 ms short: duration depends on feature interactions.

LightGBM is significantly best for duration (paired CIs). Ablation: neighbours +5.7 ms and phoneme identity +5.0 ms matter most; the pause flag is redundant with the next phoneme.

Two ways to turn timing into sound

Averaged sounds

BASELINE (AML ONLY)

Each phoneme becomes its mean spectrogram frame over the training set, repeated for its predicted duration.

Neural decoder

HYBRID (AML + DL)

Pretrained FastSpeech2: phonemes are encoded, expanded by our durations, then decoded. Output length equals the sum of our durations exactly.

Input format matters: pauses use the model's spn token and there is no word separator. A space token silently became <unk> and inflated WER from 6.9% to 20.8% on our initial test set.

A controlled, leak-free comparison

1,327 held out by SpeechBrain ∩ **1,310** in our test split → **100** evaluation sentences

SYSTEM	PAUSES	DURATIONS	SOUND
Baseline	MFA / MFA / predicted	MFA / LightGBM / LightGBM	averaged sounds
Hybrid (ours)	MFA / MFA / predicted	MFA / LightGBM / LightGBM	FastSpeech2 decoder
FastSpeech2	predicted	its own	FastSpeech2 decoder
Ceiling	real	real	real mel → vocoder

Metrics: WER (Whisper), UTMOS naturalness, duration MAE, 95% paired bootstrap CIs, blind A/B test. Fine-tuned variants of every neural system.

Same timing. Different sound.



Word error rate, lower is better. Ours and the baseline share identical text-only timing; paired difference −37.9 to −29.4 points.

Intelligibility, naturalness and listeners

TEXT-ONLY TIMING	WER [95% CI]	UTMOS
Recording	2.0% [1.2, 3.0]	4.37
Ceiling: real mel	2.2% [1.4, 3.0]	4.08
Hybrid (ours)	3.1% [2.0, 4.3]	3.70
FastSpeech2, own durations	5.3% [3.8, 7.1]	3.74
Hybrid, fine-tuned	3.6% [2.5, 5.0]	3.27
FastSpeech2, fine-tuned	5.5% [4.0, 7.4]	3.28
Baseline	36.6% [32.3, 41.1]	1.24

BLIND A/B TEST

67 / 70

judgements preferred the neural system; baseline never

Clarity 4.1 vs 1.7 out of 5, 7 listeners

Hybrid vs FastSpeech2 naturalness: −0.03 [−0.09, +0.03], not significant. A/B test ran on an earlier neural version: a conservative estimate.

Explicit timing improves the neural model

5.3%

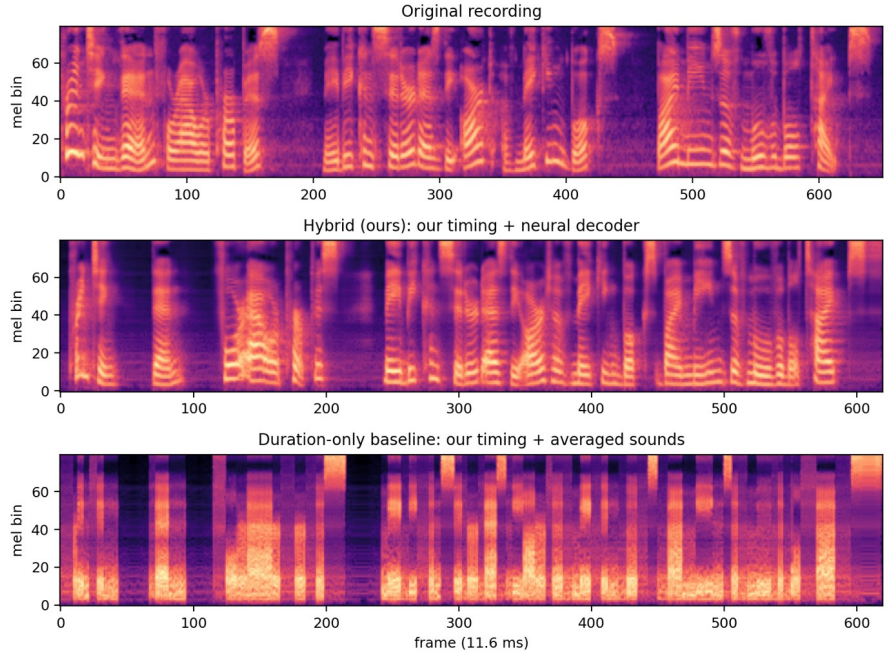
WER, model's own durations

3.1%

WER, our durations

SAME PHONEMES, PAUSES AND DECODER	OURS	FASTSPEECH2	PAIRED DIFFERENCE
WER	3.1%	5.3%	−3.6 to −0.9 pts
UTMOS	3.70	3.74	n.s.
Duration MAE	17.8 ms	23.2 ms	–

Timing without timbre



The recording and our system show curved stripes: pitch harmonics and smooth transitions.

The baseline is flat blocks. Averaging a sound across many pitches erases its harmonics.

Rhythm survives. Voice does not.

Lower loss, less natural speech

METRIC	PRETRAINED	FINE-TUNED	CHANGE
Dev mel L1	0.754	0.617	−18%
WER, raw MFA timing	4.2%	2.8%	−2.5 to −0.6 pts
WER, text-only hybrid	3.1%	3.6%	+0.1 to +1.2 pts
UTMOS, text-only hybrid	3.70	3.27	−0.48 to −0.38

Minimising average error pulls output towards an average spectrogram: over-smoothing.

Setup: decoder, output layer and postnet trained (22.5M of 43.3M params) on our split with our MFA durations; L1 loss, AdamW 5e-5, 6 epochs on a T4. Raw MFA timing has 15× more ultra-short phonemes.

Alignment decides when. Acoustics decide how.

H1	Learned acoustics are essential: 36.6% → 3.1% WER with the same timing
H2	Explicit timing improves the neural model: 5.3% → 3.1% WER, equal naturalness
H3	Fine-tuning fixes raw timing but over-smooths the voice

Next: a diffusion decoder (Grad-TTS) driven by our durations, to close the naturalness gap left by over-smoothing.

github.com/shiavm006/phonetic-guided-tts